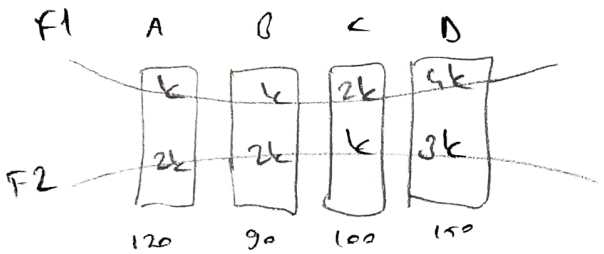
On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for this exam. The work done on this exam is totally my own. I understand by the school code, violation of these principles will lead to a zero grade and hard disciplinary issues.

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Q1)

	F1	F2	Supply Limit	cost
A	k	2k	1000	120
B	k	2k	1200	90
C	2k	k	900	100
D	4k	3k	1500	150
	200	250		



A → A1 + A2 B → B1 + B2 C → C1 + C2
D → D1 + D2

Max Z = 200F1 + 250F2 −
((A1 + 2A2) 120 + (B1 + 2B2) 90 +
(2C1 + C2) 100 + (4D1 + 3D2) 150)

constraints:
A1 + A2 ≤ 1000
B1 + B2 ≤ 1200
C1 + C2 ≤ 900
D1 + D2 ≤ 1500
F1 ≥ 200
F2 ≥ 400

Q2) Max Z = x1 + 3x2 → $x_1^+ - x_1^- + 3x_2$

x1 + x2 ≤ 2 → $x_1^+ - x_1^- + x_2 + s_1 = 2$

−x1 + x2 ≤ 4 → $-x_1^+ + x_1^- + x_2 + s_2 = 4$

x1 unrestricted → $x_1 = x_1^+ - x_1^-$

x2 ≥ 0

$x_1^+, x_1^-, x_2, s_1, s_2 \geq 0$

$x_1^+ + s_1 = 2$
$-x_1^- = 4$

$x_1^+ + x_2 = 2$
$-x_1^- + x_2 = 4$

$x_1^+ = 2$
-2

$-x_1 = 2$
x

$-x_1^- + x_2 = 2$
$x_1^- + x_2 = 4$

$-x_1^- + s_1 = 2$
$x_1^- = 4$

$-x_1^- = 2$
$x_1^- + s_2 = 4$
-2

$x_2 + s_1 = 2$
$x_2 = 4$

$x_2 = 2$
$s_1 = 2$
$s_2 = 4$

x_1^+	x_1^-	x_2	s_1	s_2	Z	f
−1	0	3	0	0		x
−4	0	0	6	0		x
2	0	0	0	6	2	✓
0	1	3	0	0	8	✓
0	4	0	6	0	−4	✓
0	−2	0	0	6		x
0	0	4	−2	0		x
0	0	2	0	2	6	✓
0	0	0	2	4	0	✓

Z = x1 + 3x2

x1 = −1
x2 = 3

